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A dataset for surface defect detection on complex structured parts based on photometric stereo

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Automated Optical Inspection (AOI) technology is crucial for industrial defect detection but struggles with shadows and surface reflectivity, resulting in false positives and missed detections, especially on non-planar parts. To address these issues, a novel defect detection technique based on deep learning and photometric stereo vision was proposed, along with the creation of the Metal Surface Defect Dataset (MSDD). The proposed Stroboscopic Illuminant Image Acquisition (SIIA) method uses a specially arranged illuminant setup and a Taylor Series Channel Mixer (TSCM) to blend multi-angle illumination images into pseudo-color images. This approach enables end-to-end defect detection using universal object detectors. The method involves mapping color space transformations to spatial domain transformations and utilizing hue randomization for data augmentation. Four object detection methods (FCOS, YOLOv5, YOLOv8, and RT-DETR) were validated on the MSDD, achieving an mAP of 86.1%, surpassing traditional methods. The MSDD includes 138,585 single-channel images and 9,239 mixed images, covering eight defect types. This dataset is essential for automated visual inspection of metal surfaces and is freely accessible for research purposes.

Background & Summary

Surface defect detection is essential in industrial production, ensuring the aesthetic quality and mechanical integrity of metal parts. Common defects such as slag inclusions, dents, cracks, insufficient casting, and pitting can significantly compromise product performance and pose serious safety risks. As reported by the NHTSA's 2022 Annual Recalls Report, over 30.8 million vehicles were recalled in the United States due to quality issues, resulting in economic losses exceeding trillions of dollars for the automotive industry[1].

Automated Optical Inspection (AOI)[2] technology has become increasingly important in detecting surface defects in metal products, offering several advantages over manual inspection, such as high efficiency, labor cost savings, and reduced susceptibility to subjective interference. The fundamental principle of AOI systems lies in utilizing the optical properties of metal surfaces. Under specific optical imaging conditions, defects exhibit distinct image features that differ from the background. 2D imaging systems and 3D imaging systems are subsequently employed to identify and localize defects or anomalies[3].

2D imaging systems. Previous research in this field has predominantly focused on defect detection using 2D imaging systems[4–8], which heavily rely on the Lambertian reflection model[9]. When light interacts with a metal surface, it produces Lambertian, directional, and specular reflections. Various lighting techniques such as bright-field, dark-field, diffuse reflection, and backlighting[10] are applied, as illustrated in Fig. 1(a-c). From these reflections, 2D images are captured, and defects are identified and located based on the contrast in brightness levels between the defects and the background. Several surface defect detection datasets have been established based on 2D imaging technologies, with the most commonly used ones being the Northeastern University Surface Defect Database (NEU-DET)[11] and the GC10-DET. These datasets have been utilized for surface defect detection using general-purpose object detection models[12–14].

While these studies have made some progress in enhancing the efficiency and accuracy of the models, they are restricted to 2D static images and lack critical advancements in optical system design, causing two main

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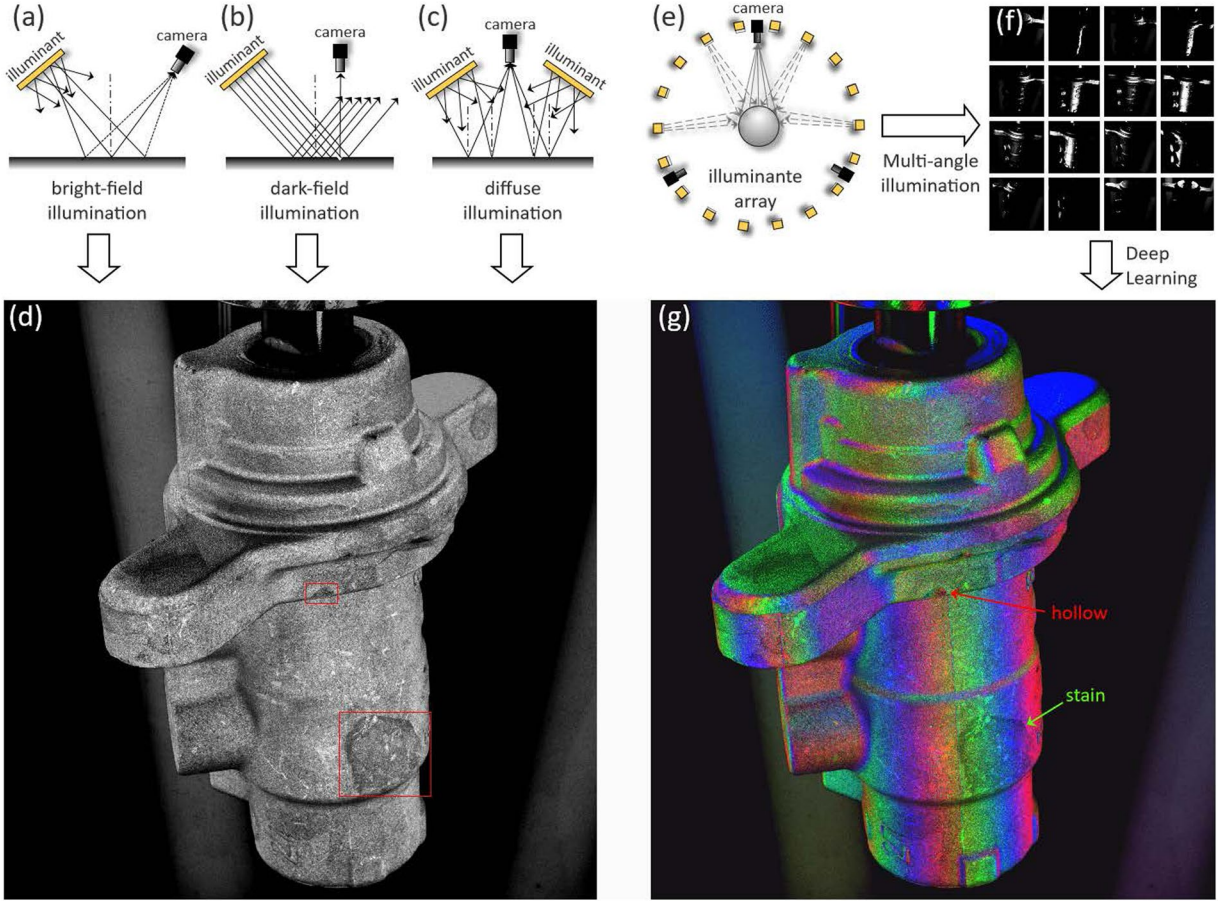


Fig. 1 Comparison between 2D imaging system and our method. Traditional methods (**a**–**c**) use bright-field illumination, dark-field illumination, and diffuse reflection illumination to obtain 2D images (**d**). Our approach utilizes an illuminate array (**e**) to capture images under multi-angle illumination (**f**). Subsequently, deep learning is employed to obtain pseudo-color images containing 3D depth information (**g**). 2D image lacks of depth information, as shown in (**d**), making it difficult to distinguish shadows from stains; Pseudo-color image with 3D depth information, as shown in (**g**), making it easier for manual detection and recognition of defects.

limitations. Firstly, their optical systems are typically highly sensitive to the angle of the lighting direction, which heavily depends on the expertise of optical engineers[15]. This dependency can lead to highly variable and less reliable detection results. Secondly, 2D images lack depth and 3D information such as normal direction, making it challenging to differentiate between shadows and stains, thereby resulting in a high false detection rate.

3D imaging systems. Therefore, 3D imaging technology has been applied in recent studies to address these challenges. 3D information is utilized to enhance the resolution of surface defect detection systems, significantly improving the resolution capability of surface defect detection systems[16], as shown in Table 1. Research indicates that different imaging modalities are complementary, and using a composite illuminant imaging system can effectively enhance the identification accuracy and stability of surface inspections[17–19]. Based on the composite illuminant imaging scheme, Woodham R. J. proposed the Photometric Stereo (PS) method[20]. This method utilizes three spatially distinct but identically illuminated illuminants to construct depth information based on the gradient of the measured surface, thereby achieving the acquisition of 3D depth image information. Other studies use structured light, stereo vision, and scanning methods for 3D defect detection[17,21,22].

Applications and challenges. These studies have achieved significant results and have been successfully applied in the production of products such as cold-rolled and hot-rolled steel strips and high-temperature continuous casting slabs. However, these investigations are still in their early stages. Most of these studies primarily focus on products with planar structures, and many significant challenges remain to be addressed for the detection of defects in parts with complex 3D structures, highlighting the necessity of advanced 3D inspection technologies:

(a) Complex structures and diverse defects. Most metal cast parts exhibit complex structures and diverse defect types that are only clearly visible under specific lighting conditions, posing challenges for 2D imaging systems. Meanwhile, 3D imaging systems are also significantly restricted by their limited working range, restricted field of view and inadequate spatial resolution, which impede their applicability in online real-time detection tasks.

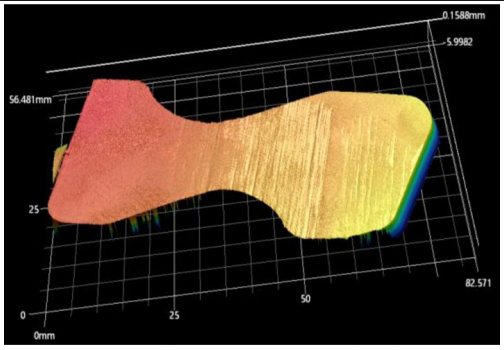
Aspect	2D Imaging Systems	3D Imaging Systems
Sample Image		
Imaging Principle	Utilizing variations in brightness levels based on Lambertian reflection models (e.g., bright-field, dark-field)	Utilizing depth information and surface gradients (e.g., photometric stereo, structured light, stereo vision)
Depth Information	Lacking the critical depth information necessary for accurate defect identification	Capturing depth information for better differentiation between defects, shadows, and stains
Field of View	Capturing a wide area, which is suitable for planar surfaces	Limited field of view, often requiring scanning and struggling with large or complex 3D structures
Surface Complexity	Suitable for flat, simple surfaces, struggling with parts that have complex geometries or diverse defect types	Suitable for more complex and curved surfaces, and better at handling diverse defect types
Speed	Faster in terms of image capture, processing, and inspection	Slower due to the need for scanning, limiting real-time detection capabilities
Real-Time Application	Suitable for real-time applications due to fast image capture and processing	Less suitable for real-time detection because scanning and reconstruction processes are slower
Cost	With lower cost for simpler and less expensive hardware	With higher cost due to complex and expensive hardware

Table 1. Comparisons between 2D and 3D Imaging Systems.

(b) Reliance on expertise. The configuration of illumination paths frequently hinges on the subjective expertise of optical engineers, which can result in highly variable and less reliable detection results.
(c) Data scarcity. There is still a scarcity of large-scale, standardized metal surface defect detection datasets. Many studies do not publicly release their datasets due to limited defect samples, category imbalances, and high annotation costs. The high hardware costs of 3D sensors make these approaches difficult to replicate, and there are no publicly available 3D imaging datasets, further limiting their widespread adoption.

Contributions of this research. To address the existing challenges in metal surface defect detection, particularly the sensitivity to lighting angles and the impact of low reflectivity on complex surfaces, this paper presents several notable contributions. These contributions focus on innovative imaging techniques, dataset creation, and comprehensive method validation. The major contributions are as follows:

(a) Development of an imaging method for non-planar parts with complex structures: This study establishes a real-time 3D imaging defect detection framework based on photometric stereo and deep learning. A Stroboscopic Illuminant Image Acquisition (SIIA) method is proposed for capturing images from light sources at all surrounding angles to address the issue of complex surface structures of parts, as shown in Fig. 1(e). By mapping color space transformations to spatial domain transformations, end-to-end optimization of the imaging system and surface defect detectors is achieved, effectively addressing sensitivity to illuminant angles and the difficulties in designing complex optical systems.
(b) Collecting and constructing of the Metal Surface Defect Detection dataset: The MSDD aluminum casting surface defect detection dataset is created, consisting of 9,332 annotated images covering eight defect types, valuable for metal part surface defect detection research. It is demonstrated that metal castings produced by the gravity casting process exhibit consistent defect patterns, making this dataset highly valuable for research in metal part surface defect detection technology.

These contributions collectively advance the field by addressing key limitations in current metal surface defect detection techniques, fostering the development of more accurate, efficient, and scalable solutions for industrial use.

Methods

The pipeline of this method consists of six steps:

(a) **Optical imaging**. Capturing images from various angles using the Strobe Illuminant Imaging Method (SIIM) for indirect 3D imaging based on photometric stereo (Fig. 1d);
(b) **Image mixing**. Combining images taken from different illuminant angles into a single RGB pseudo-color image with depth information using deep learning, by incorporating a KAN-based Taylor Series Channel Mixer for precise mixing (Fig. 1f);

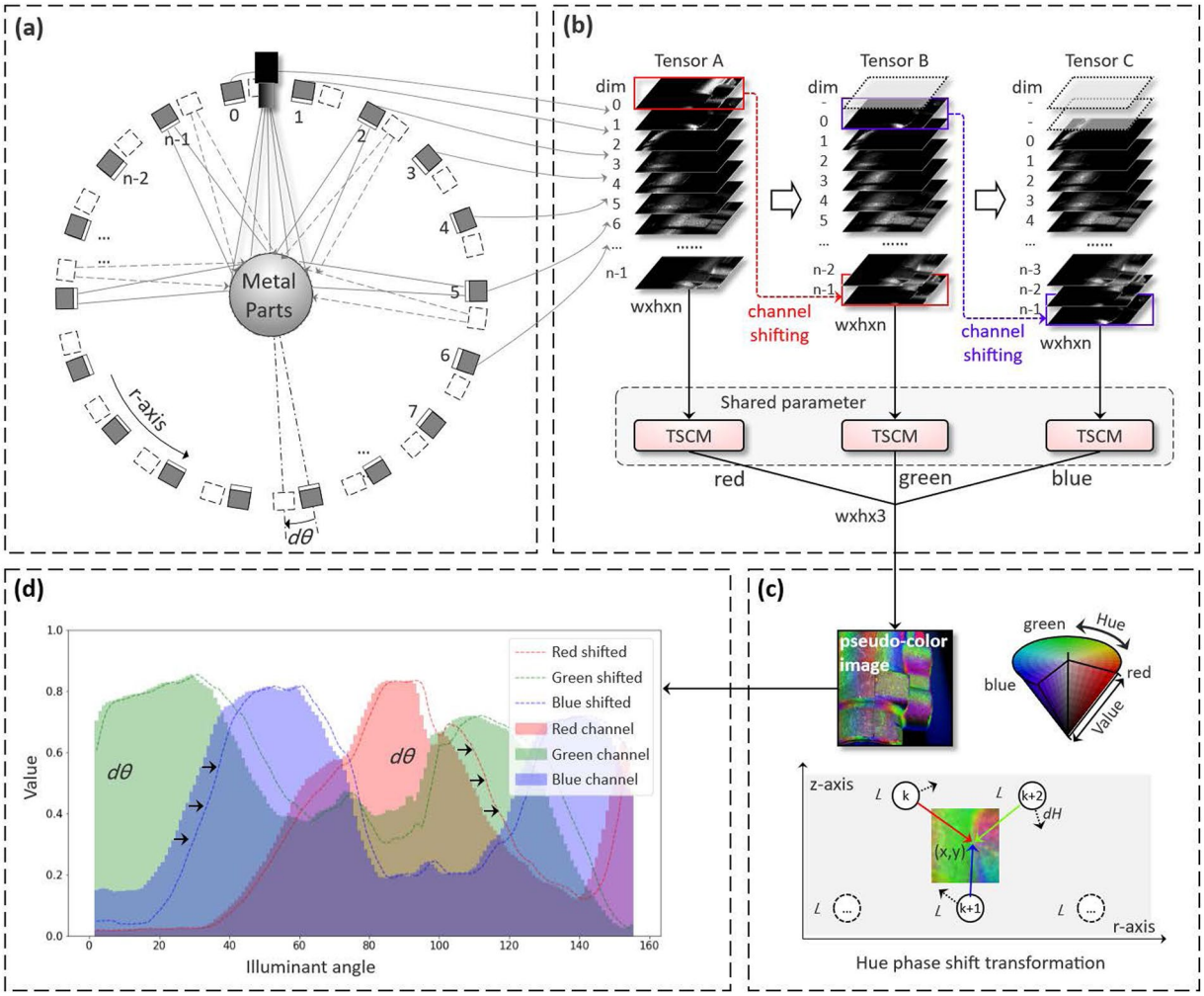


Fig. 2 Hue transformation fitting illuminant spatial transformation. Each illuminant flashes sequentially (**a**), capturing n images with different illumination angles, and stacking them into Tensor A, Tensor B and Tensor C (**b**). The three tensors are respectively input into the TSCM with shared parameters, each outputting a tensor of w × h × 1., serving as the Red, Green, and Blue channels of the mixed image to form an RGB pseudo-color image (**c**). As shown in (**c**), for any pixel (***x***, ***y***) on the metal surface, when performing a hue shift transformation on the mixed image, the hue change ***dH*** is approximately equivalent to all illuminant angles rotating by an angle ***dθ***, where $d\theta \approx T * \frac{dH}{120}$ (**d**).

(c) **Feature extraction (Optional)**. The Phase-Invariant Feature Module (PIFM) extracts lighting-invariant features by creating a stable latent space, integrating seamlessly into object detection networks by replacing early layers;
(d) **Data preprocessing**. Cutting images for annotation and training;
(e) **Data annotation**. Collecting and annotating the MSDD dataset by applying hue shift transformations to simulate lighting variations for robust feature extraction;
(f) **Defect detection**. Integrating with object detection networks for metal surface defect detection, with end-to-end optimization of optical parameters.

Optical imaging. We propose the SIIM based on photometric stereo. This method uses *m* CCD cameras and *n* illuminants arranged circumferentially around the detection object (Fig. 2a). While the schematic diagram simplifies the arrangement into a ring for clarity, in actual application, the cameras and illuminants are positioned at different heights to illuminate from various vertical and horizontal angles, capturing richer surface detail information. A programmable logic controller (PLC) alternates and synchronizes the strobing of these illuminants with the cameras, obtaining images illuminated from different angles and perspectives, as shown in Fig. 2(a). Each camera works in conjunction with the illuminants at 30 frames per second (FPS), allowing the entire image acquisition process to be completed within 0.96 seconds, which is sufficient to meet the production cycle requirements of 300 pieces per hour on a typical aluminum alloy casting line.

The classic photometric stereo method estimates the normal direction and reflectance of each point on the surface by the changes in brightness of each point under different illuminants. For each illuminant direction L_i, the pixel brightness at (x, y) can be expressed by Eq. (1):

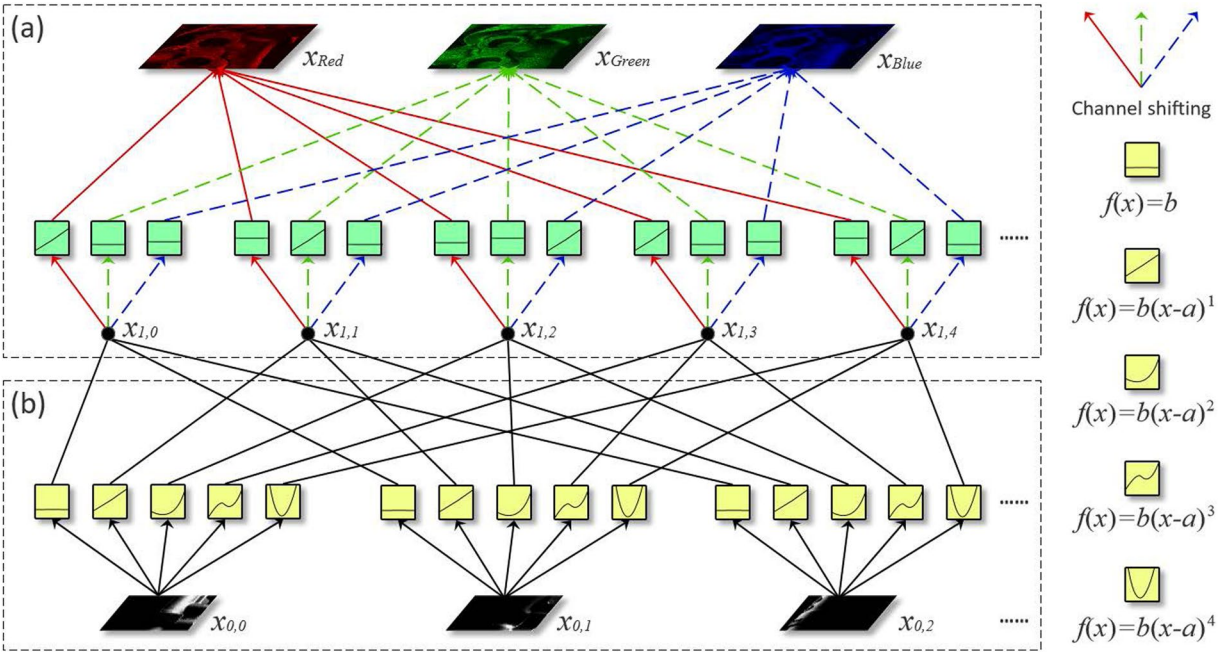


Fig. 3 Taylor series channel mixer. The multi-channel image undergoes a nonlinear activation layer (**b**), followed by a linear activation layer, which are then combined to produce the mixed three-channel image (**a**).

$$I(x, y, l) = \rho(x, y)max(n(x, y) \cdot l, 0) \tag{1}$$

Where $\rho(x, y)$ is the reflectance at (x, y), $n(x, y)$ is the surface normal vector at (x, y), and l is the direction vector of the illuminant. The n images captured are mixed in various proportion, resulting in a system of equations:

$$\begin{cases} R(x, y, l) = \Sigma\rho(x, y)\max(\theta_i n(x, y) \cdot r_i, 0) \\ G(x, y, l) = \Sigma\rho(x, y)\max(\theta_i n(x, y) \cdot g_i, 0) \\ B(x, y, l) = \Sigma\rho(x, y)\max(\theta_i n(x, y) \cdot b_i, 0) \end{cases} \tag{2}$$

Where $R(x, y, l)$, $G(x, y, l)$, and $B(x, y, l)$ are the brightness values of a pixel at (x, y) in the RGB channels, $\rho(x, y)$ is the reflectance at (x, y), $n(x, y)$ is the normal vector at that point, r_i, g_i, and b_i are the direction vectors of the illuminants in different groups, and θ_i are the parameters to be optimized for the channel mixer. By illuminating with three groups of illuminants, three equations can be obtained, thus solving for the reflectance and normal direction at point (x, y). The depth information of the surface can then be further solved using the integrated normal field or other geometric constraints methods.

Image mixing. This paper proposes a Taylor Series Channel Mixer (TSCM) that combines images from the three channels into an RGB pseudo-color image as shown in Fig. 2(c). The n illuminants are arranged in a cyclic symmetric manner, with an angle T between adjacent illuminants (Fig. 2a). First, each illuminant flashes sequentially, capturing n images with different illumination angles, denoted as 0, 1, 2, …, n-1, and stacking them into Tensor A of shape $w \times h \times n$. Next, as shown in Fig. 2(b), all channels are cyclically shifted by one position along the stacking direction to obtain Tensor B, and the shifting operation continues to obtain Tensor C. The three tensors are respectively input into the TSCM with shared parameters, each outputting a tensor of $w \times h \times 1$., serving as the Red, Green, and Blue channels of the mixed image to form an RGB pseudo-color image. Under these conditions, in the local region, for any point (x, y) on the metal surface, the positional relationship of the nearest three illuminants L_k, L_{k+1}, and L_{k+2} is as shown in Fig. 2(d), staggered in the height direction (z-axis). By optimizing the mixing parameters, the influence of illuminants farther away, such as L_{k-1}, and L_{k+3}, etc., whose angular deviation is greater than the reflective lobe range of the metal surface, on the color values at (x, y) is minimized and can thus be ignored.

The structure of the channel mixer, as illustrated in Fig. 3, includes a nonlinear activation layer and a linear activation layer. This design draws inspiration from the Kolmogorov-Arnold Network (KAN). According to the Kolmogorov-Arnold theorem, if f is a multivariate continuous function defined on a bounded domain, then f can be expressed as a finite composition of continuous functions of a single variable and the binary operation of addition. Unlike the classical KAN, which utilizes B-Splines, the channel mixer's nonlinear layer employs a fourth-order Taylor series expansion, followed by summation in the linear activation layer.

Assuming the existence of an ideal mixing function f_{mix}, it can be approximated using the Taylor series expansion formula around a point a as follows:

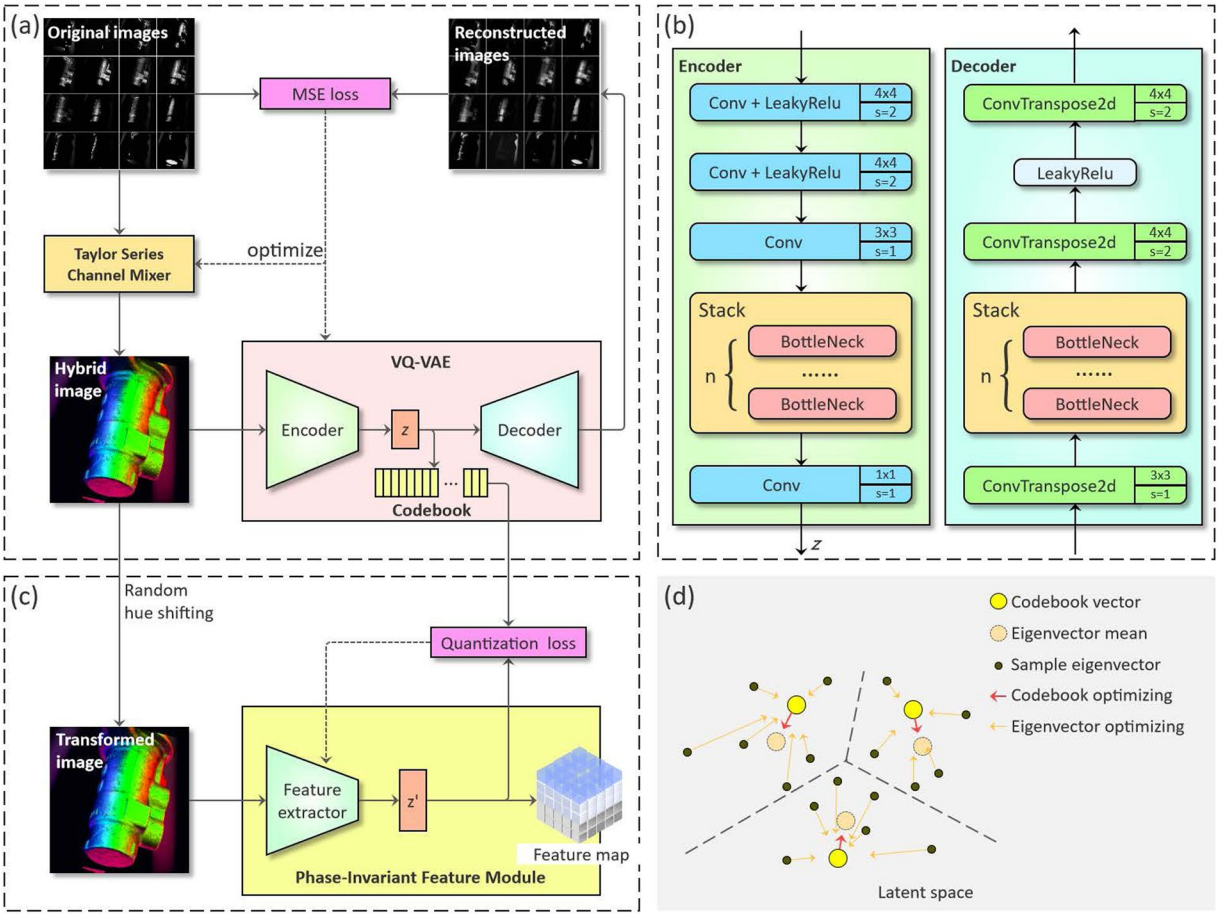


Fig. 4 Optimization of the TSCM. The Original images collected by the strobe illuminant are mixed into an RGB image through the Taylor Series Channel Mixer; The VQ-VAE optimizes the channel mixer through self-supervised learning to achieve the mixer with the least information loss (**a**); The Encoder down-samples through two convolutional layers, then obtains the encoding vector z through stacked residual blocks; The Decoder decodes z through stacked residual blocks and then up-samples through transposed convolution to restore resolution (**b**); The Codebook learns discrete representations of feature vectors in the Latent space and is optimized through Quantization loss; The MSE loss between the Original image and the Reconstructed images is used to optimize the parameters of the Taylor Series Channel Mixer and the VQ-VAE; The Phase-Invariant Feature Module learns features invariant to hue transformations by learning feature vectors before the hue transformation (**c**); Quantization loss brings the feature vectors closer to the codebook vectors in the latent space while updating the codebook vectors (**d**).

$$f_{mix}(x) \approx f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f''''(a)}{4!}(x-a)^4 \tag{3}$$

where $f(a)$ is the function value at a, $f'(a)$, $f''(a)$, $f'''(a)$ and $f''''(a)$ are the first through fourth derivatives of f_{mix} at a, and $(x-a)$ represents the deviation from a.

Given that the image pixel values are normalized between 0 and 1, this fourth-order Taylor series expansion can approximate any ideal mixing function f_{mix} with high precision within this range. Additionally, the channel mixer, consisting of only two layers, avoids issues related to gradient vanishing and explosion during backpropagation, thereby ensuring more stable network convergence compared to the Multi-Layer Perceptron (MLP) structures in traditional CNN architectures.

Under these conditions, the RGB image is transformed into the HSV space, and the hue $H(x, y)$ of each pixel (x, y) can be expressed as:

$$H(x,y) = \begin{cases} 60 \times \left(\dfrac{G' - B'}{C_{max} - C_{min}} mod\ 6 \right) & if \quad C_{max} = R' \\ 60 \times \left(\dfrac{B' - R'}{C_{max} - C_{min}} mod\ 6 \right) & if \quad C_{max} = G' \\ 60 \times \left(\dfrac{R' - G'}{C_{max} - C_{min}} mod\ 6 \right) & if \quad C_{max} = B' \end{cases} \tag{4}$$

Input: A set of parts requiring defect annotation
Output: Annotated defects for each part
for each part in the set **do**
Inspectors inspect the part manually to identify possible defects
Inspector_1 annotates defects on the part
Inspector_2 independently annotates defects on the same part
Compare annotations from Inspector_1 and Inspector_2
Initialize an empty list for final annotations
for each annotation_pair (A1, A2) from Inspector_1 and Inspector_2 **do**
Compute IoU(A1, A2)
if IoU(A1, A2) > 0.8 **then**
Compute average_box = average(A1, A2)
Add average_box to final annotations
else
Submit A1 and A2 to Quality Supervisor for review
Add reviewed_box to final annotations after approval
end if
end for
Save final annotations for the part
end for

where $C_{\min} = \max(R', G', B')$, and $C_{\min} = \min(R', G', B')$.

Regarding the overall layout, the relationship between these three tensors approximates the new images obtained by sequentially rotating the entire illuminant array by an angle $\boldsymbol{T}$ and illuminating them, while being mixed through the same channel mixer. Under this conditions, the three channels of the mixed image in the HSV color space are equivalent to a hue transformation rotated by 120 degrees. This method allows us to simulate approximately any direction of illumination conditions without moving the physical illuminants, eliminating the problem of small defects being extremely sensitive to the lighting angle.

The nonlinear layer employs a fourth-order Taylor series expansion. Because the image pixel values are normalized to the range of 0–1, this fourth-order Taylor expansion can approximate any ideal mixing function $\mathbf{f_{mix}}$ with high accuracy within this range. Additionally, compared to the MLP structure in classic CNN networks, the TSCM consists of only two layers, which helps avoid issues such as gradient vanishing and exploding during backpropagation, thereby leading to more stable network convergence.

The channel mixer is parameter-optimized using a Vector Quantized Variational Autoencoder (VQ-VAE)[23], as shown in Fig. 4. The $w \times h \times 3$. mixed image is input into the autoencoder, outputting a $w \times h \times n$. reconstructed image, and calculating the loss with the original $w \times h \times n$. image. The loss function employs Mean Squared Error Loss (MSE Loss), Structural Similarity Loss (SSIM Loss), Vector Quantizer Loss (Vector Quantizer Loss), and Commitment Loss (Commitment Loss). The MSE Loss minimizes the information loss after image mixing, while the SSIM Loss helps eliminate the "screen door" effect and detail blurring in the reconstructed image. The Vector Quantizer Loss L_{vq} makes the codebook vector e approximate the latent variable z output by the encoder, while the Commitment Loss L_{commit} makes the latent variable z output by the encoder approximate the codebook vector e. They can be expressed as:

$$L_{vq} = ||sg(z) - e||^2 \tag{5}$$

$$L_{commit} = \beta || z - sg(e) ||^2 \tag{6}$$

where sg denotes stop gradient, and β is a hyperparameter used to balance the speed at which z approximates the codebook vector and e approximates the latent variable, as shown in Fig. 4(d).

Feature extraction (Optional). This paper proposes a Phase-Invariant Feature Module (PIFM) to extract shallow feature information from images that is invariant to hue transformations, thereby eliminating the sensitivity of defect detection models to lighting angles. Notably, this step is optional. Unlike VQ-VAE, this method reconstructs images directly using the latent variable z to reduce information loss, while using e as the learning target for PIFM, as shown in Fig. 4(c). Quantization loss plays a crucial role in the autoencoder. By constructing a discrete and stable latent space, it helps the model learn features that are insensitive to subtle changes in lighting in the input image. By mapping continuous latent representations into a finite, predefined codebook space, the discrete vectors of the vectorized representation reduce the network's sensitivity to changes in lighting angles, encouraging the network to learn consistent, abstract, and conceptual information under different lighting conditions. This modular design allows the PIFM module to be easily integrated into other object detection networks. The encoder is designed as an independent feature extraction module responsible for extracting features from the image that are invariant to lighting angles, and can directly replace the P1 and P2 layers of many object detection networks, training the network using conventional methods.

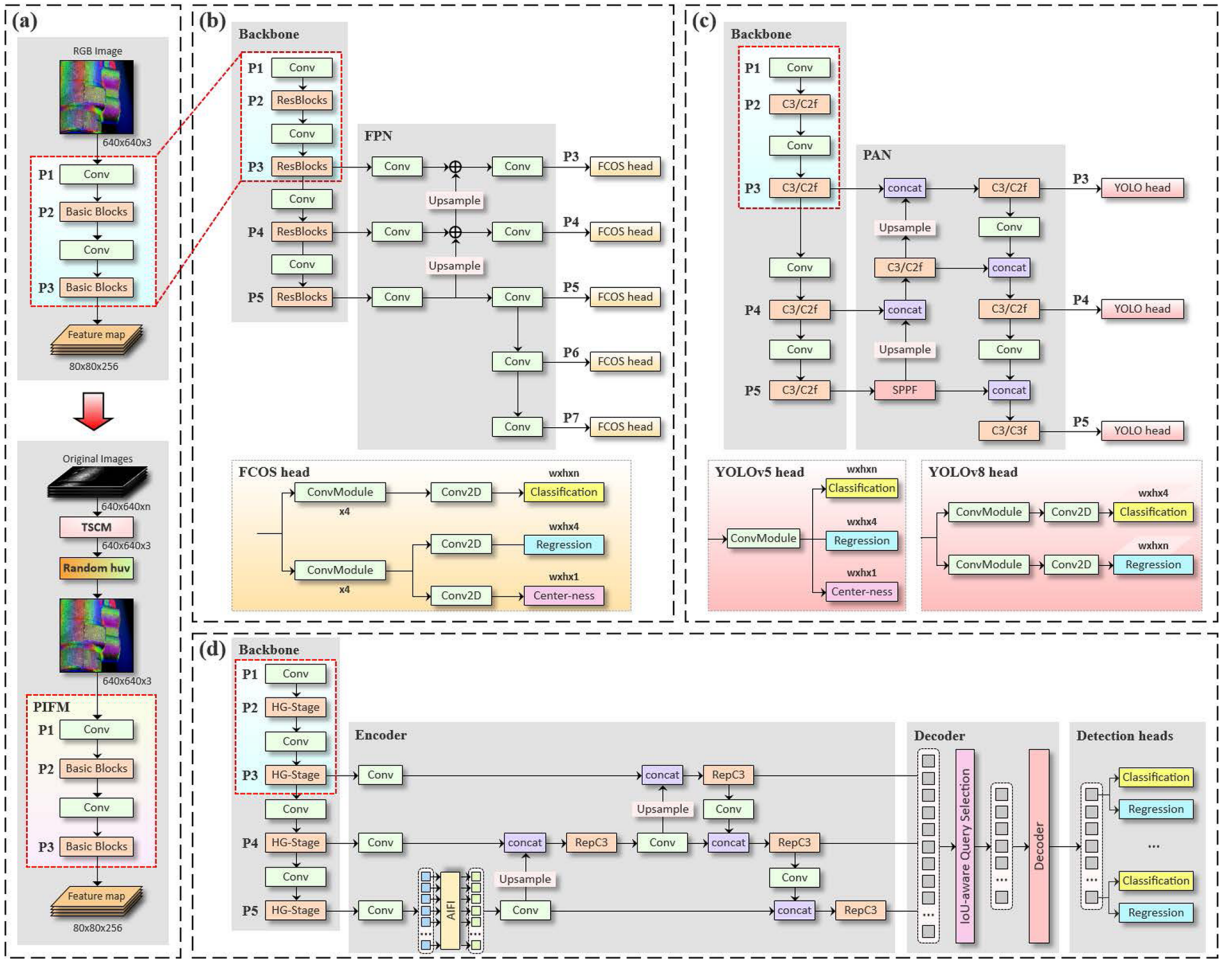


Fig. 5 Overall network architecture. (**a**) In this design, the P1-P3 stages of the backbones from conventional defect detection models is replaced by the PIFM, allowing the model to extract illumination-invariant features. This replacement enables end-to-end optimization of the TSCM, enhancing the model's robustness to lighting variations. (**b**) FCOS uses Res-Blocks in its backbone for multi-scale feature extraction, employing an anchor-free detection method. This approach reduces the complexity of anchor design and tuning, simplifying object detection while maintaining high accuracy. (**c**) YOLOv5/YOLOv8 utilize the Cross Stage Partial Block (C3 module) and CSPNet2 Modified Block (C2f module), which focuses on optimizing the separation and merging of feature maps. This design allows for more efficient computation and better feature representation by improving gradient flow and reducing redundancy in feature extraction. (**d**) RT-DETR incorporates the Hourglass Block (HG-Block) for effective multi-scale feature extraction and spatial context aggregation. Furthermore, it leverages a transformer architecture, allowing the model to capture global dependencies within images.

Data preprocessing. The dataset collected 1,507 metal parts with defects. Images were captured multiple times under different lighting conditions by six cameras, resulting in an initially large dataset. This method utilizes high-resolution cameras (up to 2560 × 1920 pixels) to detect defects as small as 0.5 mm. These images enable the model to effectively identify tiny defect features. To manage large image sizes, a sliding window technique divides images into 640 × 640 patches, with a 15% overlap to ensure complete defect detection. All images were slightly distorted, scaled, and sheared to obscure the true dimensions and shapes of the products. Additionally, all logos and identifying marks were removed to prevent potential commercial disputes. Experimental results indicate that these modifications did not significantly affect defect detection performance.

Data annotation. The data annotation process was completed by two professionally trained inspectors and one quality supervisor. Before starting, the inspectors manually confirmed the defects on the parts to avoid any influence from optical enhancement methods. Then, the two inspectors collected images of the parts and independently annotated the enhanced images. After both inspectors completed their annotations, the IoU of the annotation boxes was used to determine whether a review by the quality supervisor was needed. If a review was necessary, the quality supervisor finalized the annotations; otherwise, the average of the two annotation boxes was taken as the final result. The annotation process can be represented by the following pseudocode:

Defect detection. The images collected in this study can be utilized for training and detection in general object detection models by simply inputting the TSCM-mixed RGB pseudo-color images into these networks, similar to how other defect detection datasets are used. To incorporate the PIFM and enable end-to-end optimization, the P1-P3 stages of the backbones in conventional defect detection models can be replaced with the PIFM, as illustrated in Fig. 5a. This modification is independent of the object detection model architecture, whether it is anchor-free, anchor-based, or ViT-based. Accordingly, we have tested this approach on FCOS[24] (Fig. 5b), YOLOv5[25], YOLOv8[26] (Fig. 5c), and RT-DETR[27] (Fig. 5d). The Ultralytics[26] framework was utilized to facilitate the rapid construction of object detection networks, although it is not essential to our method.

The fully modular design endows this network with robust generalization capabilities, enabling the replacement of the object detector with any standard object detection network architecture. The encoder is responsible for extracting features from the image that are invariant to lighting angles, and can directly replace the P1-P3 layers of many object detection networks, training the network using conventional methods. By pre-training the model's feature extraction layers in this way, convergence speed and detection accuracy can be improved. The hue transformation is entirely implemented on tensors to maintain gradient flow, making the proposed method end-to-end optimizable, allowing fine-tuning of the channel mixer and feature extractor even in the final stages of defect detection. Compared to static mixing methods, this approach not only addresses the sensitivity to illuminant angles but also provides stronger feature representation capabilities.

Data Records

The entire dataset is stored in the 'datasets' folder, which includes three subfolders: 'OrigImages', containing images with multi-angle illumination, 'JPEGImages', containing the mixed pseudo-color images, and 'Annotations', containing the annotation files. The dataset[28] has been uploaded to the Science Data Bank and can be accessed at https://doi.org/10.57760/sciencedb.10794.

Defect categories definitions. There are eight types of defects, namely Misrun, Inclusion, Dent, Parting line crack, Stamp collapse, Pockmarks, Mould scuffing, and Cut marks. The definitions of the defect categories are provided in Table 2.

Dataset structure. The folder structure of the dataset is shown in Fig. 6. The dataset consists of image files and annotation files, stored in the Images and Annotations folders respectively. Each image has a corresponding annotation file, with filenames containing the date, time, and device ID of the image capture. The annotation files use the VOC format and include the filename, dimensions, and the category and bounding box location of each defect in the image.

Technical Validation

The experiments were conducted on a computer with an Intel Xeon(R) Silver 4110 CPU @ 2.10 GHz processor, 32 GB memory, 24 GB NVIDIA GeForce 3090 GPU, and Windows 10 operating system. The network was built using the Pytorch 1.10.1 framework. The dataset used was the MSDD dataset, split into training and testing sets in a 4:1 ratio.

To verify the performance improvement of the proposed method, ablation experiments were conducted using various improved methods to evaluate their impact on the model's performance. As mentioned earlier, the improvements mainly include:

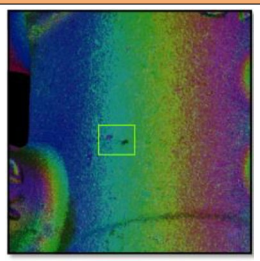
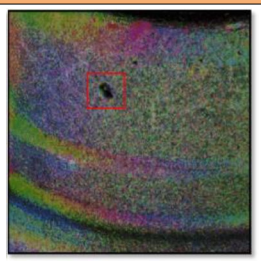
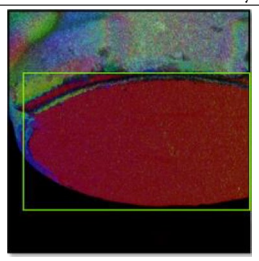
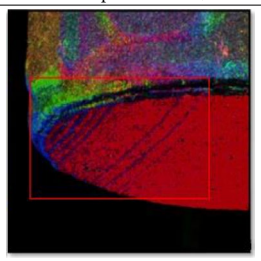
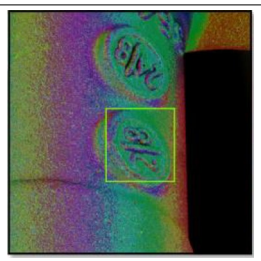
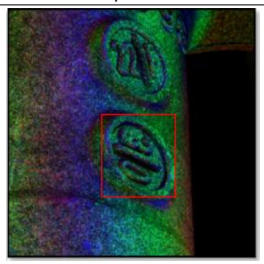
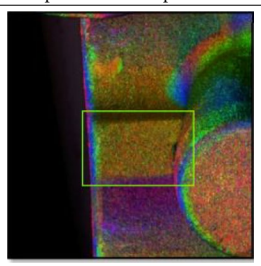
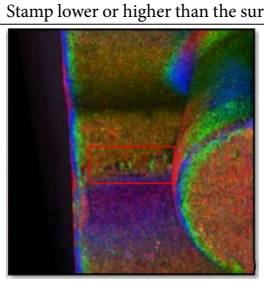
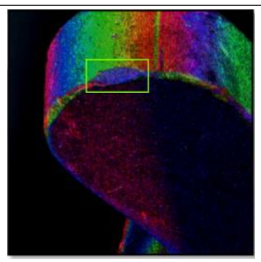
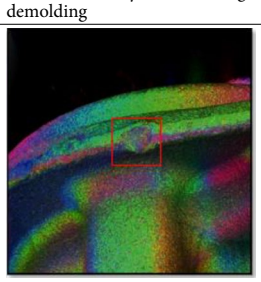
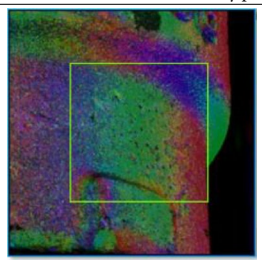
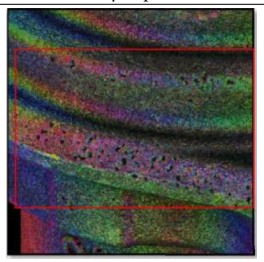
(a) Linear channel mixer;
(b) Adding a nonlinear expression branch utilizing CNN and ReLU to give the channel mixer nonlinear expression capability;
(c) Using TSCM;
(d) Using a Vector Quantization layer to construct a discrete latent space, and applying random hue phase shift transformations to make the defect detection network insensitive to subtle changes in illuminant direction, and improving the robustness of the detection model.

The ablation experiments mainly compared the reconstruction image errors of different mixing methods on the test set, including Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and mean Average Precision (mAP) at IoU threshold 0.50 (mAP_{50}) for defect detection.

The results of the ablation experiments are shown in Table 3. The results indicate that the TSCM improved the PSNR by 6.02 to 6.03. The vector quantization layer had an insignificant impact on reconstruction quality but increased the mean Average Precision for defect detection by 1.6%. The former shows blurred details and some information loss, while the latter produces reconstructed images almost indistinguishable from the original images to the naked eye.

To further verify the effectiveness of this method, we compared it with state-of-the-art general object detection networks, including FCOS, YOLOv5, YOLOv8, and RT-DETR. The baseline version used a static image mixing method, with mixing parameters optimized by the autoencoder; the improved version used an end-to-end optimized image mixing method with gradient-based random phase shift transformations for data augmentation. Additionally, as a control, the model input channels were modified to n, and the n-channel original images were directly input into the model for training without mixing.

The experimental results are shown in Table 4. Due to the small sample sizes of misrun and pockmarks defects, which are not statistically significant, the experiment only compared five types of defects: dent, inclusion, parting line crack, stamp collapse, and mould scuffing. The results indicate that the method of defect detection using static mixed RGB images significantly outperformed the direct training without mixing. Regardless

Defect Categories	Defect Definition	Comparison of Non-Defects and Defects	
		Non-defect	Defect
Inclusion	Internal or surface defects caused by non-metallic impurities in the metal melt. Depth > 0.5 mm		
		Surface stain with low reflectivity	Hole with depth > 0.5mm, inclusion defect;
Cut marks	Knife marks or scratches caused by improper clamping of the part or unstable movement of the cutting tool during riser cutting. Depth > 0.1 mm		
		Smooth surface	Lines caused by tool vibration during cutting
Stamp collapse	Local collapse on the casting surface caused by deep stamping of dates or marks on the mold. Depth or height > 0.5 mm		
		Stamp flush with the part surface	Stamp lower or higher than the surface
Mould scuffing	Local defects on the casting caused by uneven distribution of mold release agent, leading to metal sticking to the mold surface during demolding. Depth > 0.5 mm		
		Normal demolded surface	Defects caused by metal sticking to the mold during demolding
Dent	Surface depressions caused by impacts on the casting before heat treatment. Depth > 0.5 mm		
		Normal deformation caused by part	Dent caused by impact before heat treatment
Pockmarks	Numerous small pits or round depressions on the casting surface, usually caused by rapid local cooling or high hydrogen content in the aluminum melt. Depth > 0.5 mm		
		Numerous but shallow small holes	Numerous small deep holes

Continued

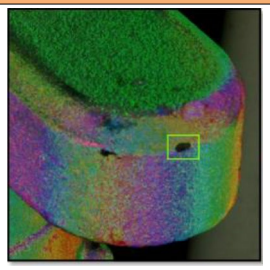
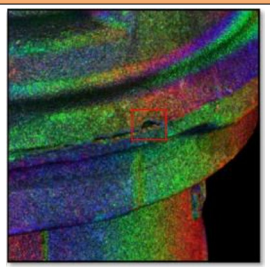
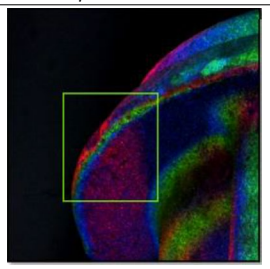
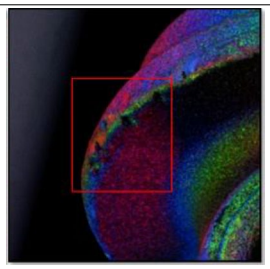
Defect Categories	Defect Definition	Comparison of Non-Defects and Defects	
		Non-defect	Defect
Parting line crack	Irregular gaps at the mold parting line due to thin metal layers adhering to the mold surface. Depth > 0.5 mm		
		Stain at the parting line with low reflectivity	Deep hole or crack at the parting line
Misrun	Incomplete part shape due to molten metal failing to fully fill the mold. Depth > 0.5 mm		
		Normal random deformation at the part edge	Defect at the part edge due to insufficient molten aluminum temperature

Table 2. Defect category definitions[29].

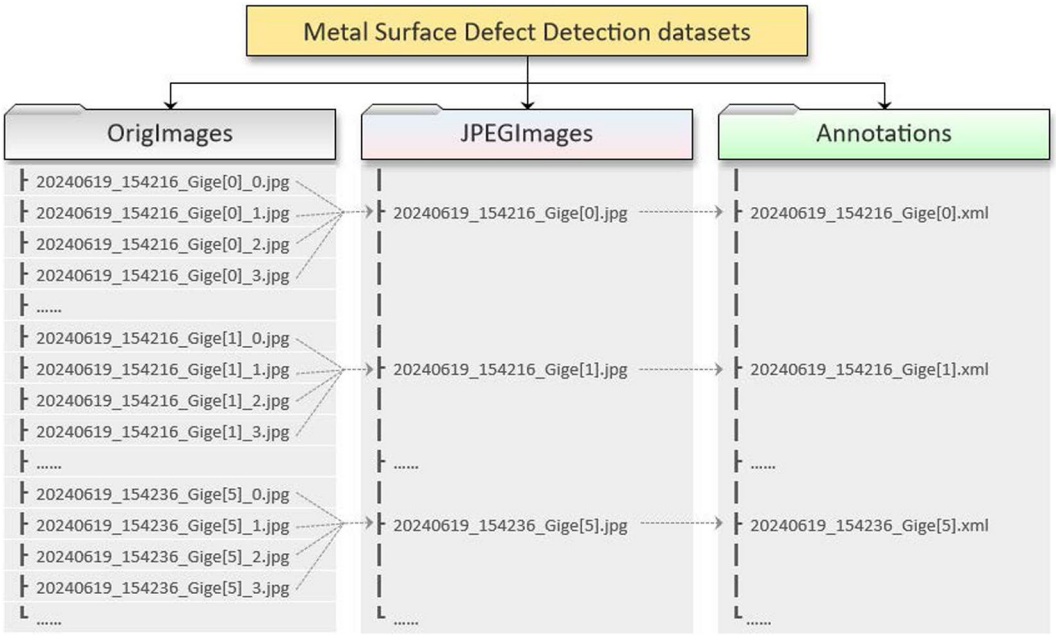


Fig. 6 Dataset structure.

of the object detection network used, the PIFM module implemented with phase shift transformations brought an average precision improvement of 4.1% to 5.5% compared to the static mixing scheme. The end-to-end optimized channel mixer provided an additional average precision improvement of 1.1% to 2.1%. The YOLOv8-l network performed best on the MSDDs dataset, achieving mAP_{50} of 0.854, as shown in Fig. 7. The accuracy of the YOLOv8-m network was almost identical to that of YOLOv8-l but with faster detection speed, making it the best overall performer.

From the perspective of specific defect categories, the phase shift transformation had a minimal impact on the defects of stamp collapse and parting line crack, only improving the average precision by about 1%. This is likely because these defects occur in relatively fixed positions and are less sensitive to illuminant angles. However, the dent and inclusion defects, which have variable positions on the parts and are greatly affected by

Linear Mixer	CNN + ReLU	TSCM	Vector Quantization	MSE_{test}	$PSNR_{test}$	$SSIM_{test}$	mAP_{50}
√				0.00071	31.49	0.0782	0.759
	√			0.00062	32.21	0.0760	0.765
		√		0.00015	38.24	0.0299	0.768
√			√	0.00061	32.15	0.0782	0.777
	√		√	0.00056	32.52	0.0716	0.780
		√	√	0.00014	38.54	0.0287	0.784

Table 3. Ablation experiment result.

Models	Static Mixing	TSCM	PIFM	End-to-End Optimization	mAP_{50}					
					All Classes	Dent	Inclusion	Parting Line Crack	Stamp Collapse	Scraping
FCOS + ATSS	√				—	—	—	—	—	—
		√			0.628	0.318	0.677	0.695	0.923	0.748
		√	√		0.669	0.379	0.769	0.702	0.940	0.742
		√	√	√	0.680	0.453	0.756	0.706	0.950	0.763
YOLOv5-m6	√				0.704	0.604	0.800	0.770	0.931	0.790
		√			0.774	0.533	0.827	0.792	0.968	0.881
		√	√		0.819	0.650	0.869	0.806	0.972	0.845
		√	√	√	0.840	0.696	0.853	0.834	0.97	0.877
YOLOv8-m	√				0.722	0.516	0.743	0.766	0.941	0.792
		√			0.783	0.508	0.791	0.802	0.966	0.857
		√	√		0.838	0.664	0.867	0.845	0.970	0.867
		√	√	√	0.853	0.671	0.87	0.847	0.975	0.890
YOLOv8-l	√				0.748	0.693	0.574	0.590	0.980	0.846
		√			0.783	0.508	0.791	0.802	0.926	0.857
		√	√		0.847	0.723	0.872	0.819	0.978	0.892
		√	√	√	0.861	0.708	0.874	0.823	0.977	0.940
RT-DETR-l	√				0.711	0.496	0.763	0.773	0.943	0.781
		√			0.759	0.523	0.807	0.777	0.964	0.840
		√	√		0.799	0.523	0.831	0.854	0.956	0.815
		√	√	√	0.815	0.532	0.807	0.787	0.967	0.841

Table 4. Comparison of object detection model performance: Static Mixing, TSCM, PIFM and End-to-End Optimization.

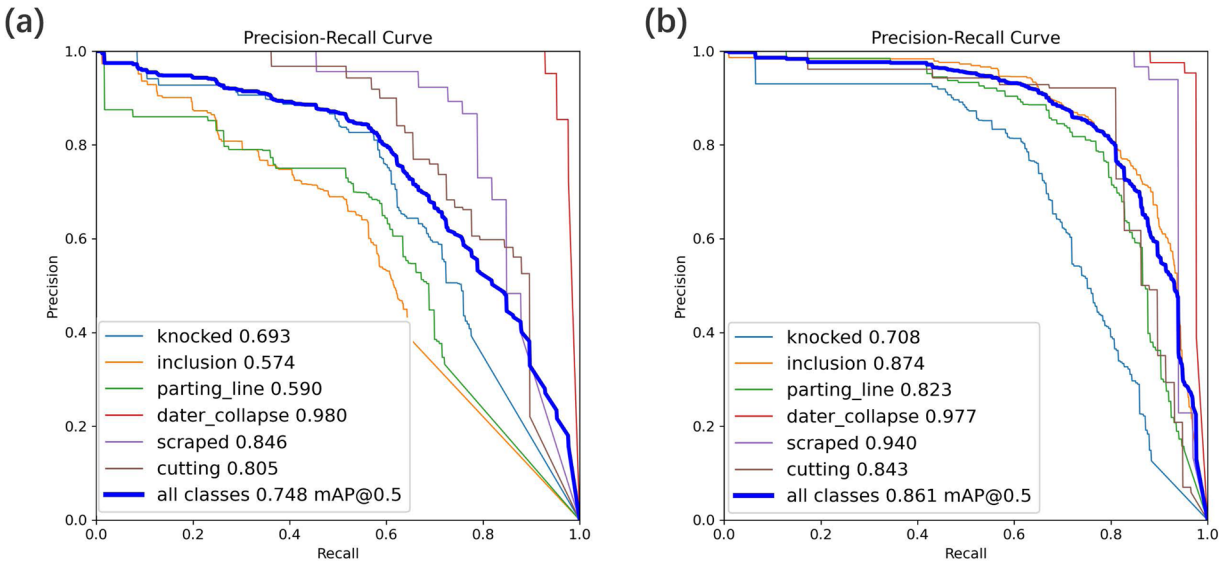


Fig. 7 Comparison of detection performance. (**a**) YOLOv8-l with Static Mixing; (**b**) YOLOv8-l with Phase Shift Transformation and End-to-End Optimization.

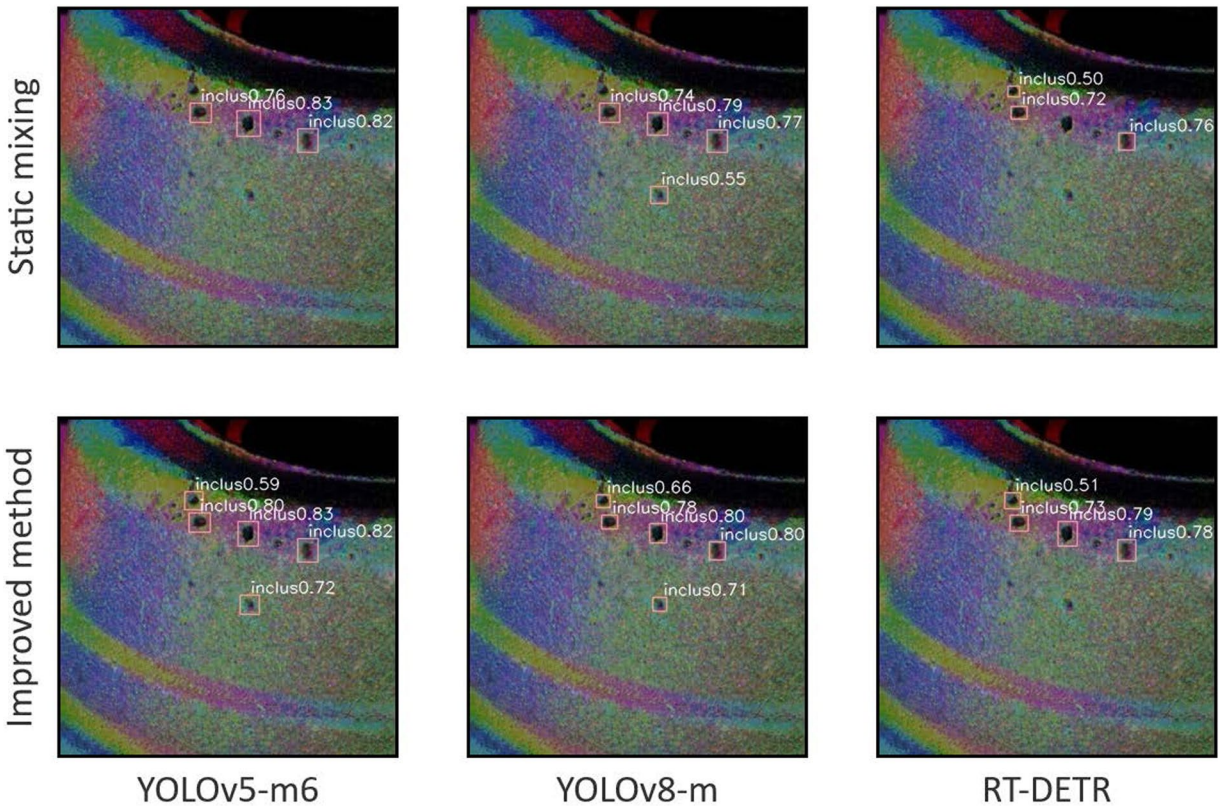


Fig. 8 Comparison of Detection Results between Static Mixing and Improved Methods. For the best three models (YOLOv5-m6, YOLOv8-m, and RT-DETR), the detection performance of the Improved method is significantly better than that of the Static mixing method.

illuminant angles, saw an increase of up to 18.7% in average precision, with significant improvements in accuracy and recall rates, as shown in Fig. 8.

The complex geometric shapes of the materials cause 100% of the samples to be affected by shadows and occlusions, significantly impacting detection performance. Additionally, low reflectivity caused by stains is challenging to quantify since stains, as non-defects, are not annotated. However, as aluminum alloy casting blanks are often stored outdoors, they are exposed to environmental factors such as rain and hurricanes, leading to batch occurrences of stains, which contributes to a significant number of false detections in real-world settings.

To address these challenges, the proposed method integrates photometric stereo-based strobe imaging and deep learning to enhance metal surface defect detection. Its effectiveness is validated using the high-quality MSDD dataset. As shown in Table 5, compared to NEU-DET and GC10-DET, the MSDD dataset includes not only a larger collection of grayscale images but also RGB pseudo-color images with depth information, adding an extra layer of detail for more accurate detection. This makes the MSDD dataset an essential tool for addressing surface defect detection on non-planar materials, a critical challenge in industrial applications.

The key elements of the proposed method to overcome these challenges are outlined below:

(a) **Imaging and illumination**. The Stroboscopic Illuminant Image Acquisition (SIIA) method captures images from multiple angles to address complex surface geometries, combining them into RGB pseudo-color images with depth information using the Taylor Series Channel Mixer (TSCM). This method enhances robustness to varying lighting conditions.
(b) **Dataset quality**. The MSDD dataset, with 9,332 annotated images covering eight defect types, ensures high model performance. The dataset's diverse coverage of gravity-cast metal part defects enables effective model training for various industrial applications.
(c) **Scalability**. The method has demonstrated adaptability across different model architectures, including anchor-free, anchor-based, and ViT-based models. These general object detection models exhibit superior performance on the MSDD dataset compared to traditional static lighting approaches, highlighting the method's flexibility. This shows that the method is independent of the underlying model architecture, making it scalable and suitable for detecting defects on surfaces with complex geometries and under various lighting conditions. This flexibility allows it to be integrated across different industrial settings, broadening its application potential.

Figure 9 shows the comparison of effects between traditional static ring illumination and photometric stereo imaging. Achieving globally uniform diffuse reflection illumination with static ring light makes it difficult to

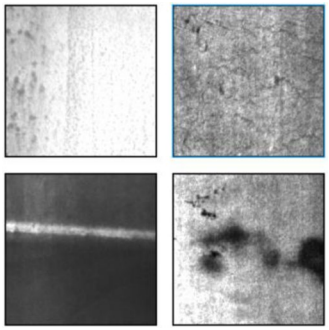
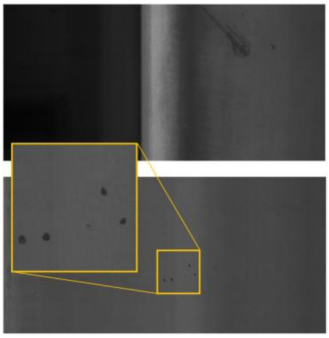
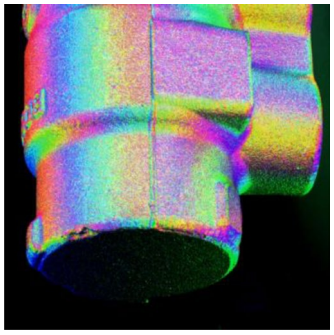
Aspect	NEU-DET	GC10-DET	MSDD
Sample Image			
Gray Images	1800	2280	149312
RGB Images	—	—	9332
Defects Type	Crazing, Inclusion, Patches, Pitted Surface, Rolled-in Scale, Scratches	Crack, Welding Line, Inclusion, Scratch, Pit, Bump, Break, Rolled-in Scale, Oil Spot, Dirty Spot	Inclusion, Cut marks, Stamp collapse, Mould scuffing, Dent, Pockmarks, Parting line crack, Misrun
Image Size	200 × 200	2048 × 1000	640 × 640
Structural Complexity	Plane	Plane	Non-plane

Table 5. Comparison of the NEU-DET, GC10-DET and MSDD.

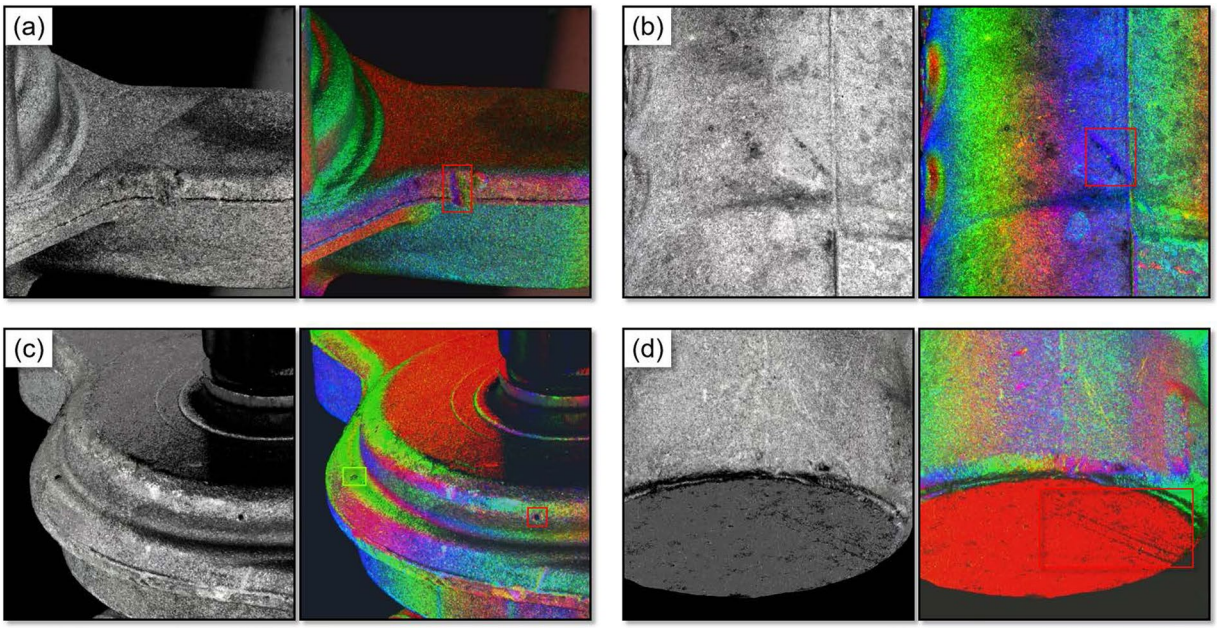


Fig. 9 Comparison between 2D imaging system and our method. (**a**) Left: The concave shapes are not clearly visible; Right: The concave shapes are clearly delineated using our method; (**b**) Left: Distinguishing between shadows and stains is challenging in traditional methods; Right: It is easier in the pseudo-color image; (**c**) Left: In the 2D image, two black spots are visible; Right: Only one is an inclusion (red box), while the other is a shadow caused by a protrusion (green box); (**d**) Left: The diffuse lighting source reduces the contrast of cut marks; Right: In the pseudo-color image, cut marks are more prominently featured.

distinguish between low reflectivity caused by surface stains and shadow areas; whereas in the pseudo-color image, the local concave shape of the defect is very clear, significantly improving the recall rate for both manual inspection and defect detection models.

Limitations. While our method has shown strong performance, there are several limitations that should be noted:

(a) SIIA method on reflective surfaces. One of the key challenges we encountered is that the SIIA (Strobe Imaging Illumination Adjustment) method struggles with highly reflective surfaces. The complex arrangement of light sources tends to produce clear reflections on smooth surfaces, which can interfere with defect

detection by creating visual artifacts that obscure the defects themselves. This is a significant issue when applying the model to surfaces like polished metals or glass.

(b) Imbalanced data. Another limitation involves the issue of data imbalance in our dataset, which could impact the model's ability to generalize across various defect types. This is particularly evident when certain defect categories are underrepresented, potentially leading to skewed results or lower detection rates for rarer defects.

Future research directions. To improve the model and bring it closer to industrial application, several future research directions are worth exploring:

(a) Addressing imbalanced data. Continued data collection is essential to address the issue of imbalanced data categories. Expanding the dataset to include more diverse samples will improve the model's ability to handle underrepresented defect types, leading to more robust performance across a variety of industrial scenarios.

(b) Testing on non-metallic surfaces. While initial experiments have demonstrated that the method works well on non-metallic surfaces, such as wood, leather, and fabric, showing promising extensibility, further experimentation and validation are necessary to confirm these findings across a wider variety of non-metallic materials.

(c) Improving detection accuracy. Additional efforts are required to improve the model's detection accuracy to meet the stringent standards demanded by industrial environments. This can be achieved by optimizing the network architecture or incorporating more diverse training data to refine the model's performance.

Usage Notes

In actual production, the majority of parts are defect-free, yet it is common to erroneously classify qualified parts as defective. Therefore, this dataset includes a certain number of background images without defects to help the model learn the patterns of normal parts. When training the dataset using general object detection models, Translate, Scale, Mosaic, Random-Affine, MixUp, Augment HSV, and Horizontal Flip have been verified as effective data augmentation methods. Additionally, unlike general object detection tasks, the features of surface defects are highly correlated with the features of their surrounding background. Therefore, the positive and negative sample labeling strategy has a significant impact on the performance of the detection model.

Code availability

The code used for technical validation in this paper is available at https://github.com/Destinyia/Metal-Surface-Defect-Detection. The code follows the Apache-2.0 license and can be accessed freely for non-commercial use. We provide demo scripts for key steps: 'train_tscm.py' demonstrates training the TSCM, 'train_pifm.py' is used for pre-training the PIFM, and 'train_detector.py' shows the training process of the defect detector. Additionally, partial training results are retained in the 'runs' directory for reference.

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Author contributions

Lin Wu and Yu Ran are co-first authors of this paper. Lin Wu completed the writing of the paper, designed the method, implemented the code, and conducted the technical validation. Yu Ran contributed to the organization and creation of the dataset and provided valuable suggestions for the writing of the paper. Li Yan helped with the grammatical revision and language polishing. Professor Dongran Han and Professor You Song, as the supervising teacher, guided the research work and conducted the final review of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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